

Santushti Singh

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EDUCATION

Techno India University

2022 - 2026

Btech, Computer Science Engineering AIML

Current GPA: 8.19

Class XII: 86% Class X: 76%

SKILLS

Languages: C/C++, Python, Java, JavaScript, TypeScript

Framework: React.js, React Native, REST APIs, Node.js, Express.js, Django, Django REST Framework, FastAPI

Tools: Git/GitHub, Figma, Arduino, ESP32, Postman

PROJECTS

Battery Energy Storage & Solar PV Decision Support System

- Built a React.js/TypeScript web app for renewable energy planning and data – driven decision support
- Implemented calculation workflow for load estimation, BESS sizing, solar PV sizing, CAPEX, cash flow, NPV, IRR, Payback year.
- Integrated database – backend calculations API – based application workflows

Virtual Try-On Platform (Ongoing)

- Currently developing a virtual try-on web application that enables users to visualize clothing on a person using image-based processing and AI Models
- Developing an AI-powered virtual try-on platform using React Native, Python, FastAPI and CatVTON diffusion models, with automated image preprocessing and clothing-mask generation.

Petition Analysis & Management Platform — Initial Petition

- Developed a full-stack web application for petition management and automated petition analysis, enabling users to interact with petition content through a web – based interface
- Built a python Flask backend with REST API endpoints for health monitoring, PDF upload, text extraction, and petition analysis

Customer Behaviour Analysis

- Performed exploratory data analysis using Python and SQL to identify customer segments and purchasing patterns
- Used Numpy and Pandas for data manipulation, cleaning, transformation, and preprocessing.
- Used SQL queries to retrieve, filter, aggregate and analyse relational data

IoT Water Quality Monitoring System

- Developed a smart water quality monitoring system using ESP32 for real-time data transmission.
- Integrated TDS sensor, TDO sensor, and pH sensor to measure water quality parameters. Transmitted sensor data to a database via Wi-Fi for storage and analysis.
- Designed a frontend dashboard for real-time monitoring and visualization. Developed REST APIs using Node.js and Express.js and stored data in MongoDB. Designed and deployed a real-time dashboard for monitoring and visualization

EXPERIENCE

Research Analyst/ Fullstack Developer Intern (IRADe)

May 2026 – Aug 2026

- Developed a production-oriented React.js/TypeScript web application for renewable energy planning and data-driven decision support, financial analysis modules and an interactive user
- Developed an automated SMTP –based system capable of sending bulk emails to thousands of recipients per run using python

FrontendIntern — Vaultfcodes

May 2025- June 2025

- Collaborated with a development team to build, test, and enhance responsive, clean UI components.

CERTIFICATIONS

- Java (NPTEL, 12-week): Java fundamentals, data structures, OOP, JDBC (database connectivity).
- Smart India Hackathon (SIH): Selected participant; developed solutions for real-world problem statements